

Regime	FedAvg	FedProx	FedNova	Key mechanism
IID / near-IID	competitive	no gain	competitive	no heterogeneity to separate
Statistical heterogeneity only	competitive	fragile gain	not tested	uneven rare-class ownership; small per-class rebalancing
L4 straggler (rare-client signal)	collapse	rescue (γ -inex)	not tested	γ -inexact admits the rare client's dropped partial update
Deterministic LR asymmetry	collapse	collapse	robust	τ -normalisation rescales the weakened rare client
Random-τ stragglers	competitive	competitive	collapse	τ -normalisation amplifies noisy partial updates
Loss-side correction (L4)	competitive	no extra gain	not tested	weighted CE re-weights rare classes (a distinct layer)


 preserved / competitive / rescue / robust


 fragile / no extra gain


 collapse / brittle


 not tested in this thesis